

Theme 4 Definitions

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Theme 4

4.1 Max and Min

- 1 Definitio** Let c be a number in the domain D of a function f . Then $f(c)$ is the
- **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all x in D .
 - **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all x in D .

- 2 Definitio** The number $f(c)$ is a
- **local maximum** value of f if $f(c) \geq f(x)$ when x is near c .
 - **local minimum** value of f if $f(c) \leq f(x)$ when x is near c .

- 6 Definitio** A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

- 3 The Extreme Value Theorem** If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

- 4 Fermat's Theorem** If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

The Closed Interval Method To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b) .
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

4.2 MVT and Rolle's

Rolle's Theorem Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$

Then there is a number c in (a, b) such that $f'(c) = 0$.

The Mean Value Theorem Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

Then there is a number c in (a, b) such that

$$\boxed{1} \quad f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

$$\boxed{2} \quad f(b) - f(a) = f'(c)(b - a)$$

4.3 Using Derivatives

Increasing/Decreasing Test

- (a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.
- (b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

The First Derivative Test Suppose that c is a critical number of a continuous function f .

- (a) If f' changes from positive to negative at c , then f has a local maximum at c .
- (b) If f' changes from negative to positive at c , then f has a local minimum at c .
- (c) If f' is positive to the left and right of c , or negative to the left and right of c , then f has no local maximum or minimum at c .

A function f is called **increasing** on an interval I if

$$f(x_1) < f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

It is called **decreasing** on I if

$$f(x_1) > f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

IMPORTANT!

Definitio If the graph of f lies above all of its tangents on an interval I , then f is called **concave upward** on I . If the graph of f lies below all of its tangents on I , then f is called **concave downward** on I .

Concavity Test

- (a) If $f''(x) > 0$ on an interval I , then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ on an interval I , then the graph of f is concave downward on I .

Definitio A point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P .

The Second Derivative Test Suppose f'' is continuous near c .

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

4.4 Indeterminate Forms

L'Hospital's Rule Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or that $\lim_{x \rightarrow a} f(x) = \pm\infty$ and $\lim_{x \rightarrow a} g(x) = \pm\infty$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.) Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

7 Forms :

$$\frac{0}{0}, \frac{\pm\infty}{\pm\infty} : \text{L'Hospital's}$$

$$0 \cdot \infty :$$

$$f \cdot g = \frac{f}{\frac{1}{g}} \quad \text{or} \quad \frac{g}{\frac{1}{f}}$$

$$\infty - \infty :$$

find common
denominator

$$0^0, \infty^0, 1^\infty :$$

$$y = [f(x)]^{g(x)}$$

$$\ln y = g(x) \ln[f(x)]$$

$\mathbb{O} \mathbb{R}$

$$[f(x)]^{g(x)} = e^{g(x) \ln f(x)}$$

Remark

You always have to write down when a limit has an indeterminate form and you have to indicate that you use L'Hospital's rule, e.g.,

$$\begin{aligned} & \lim_{x \rightarrow \infty} \frac{e^x}{x^2} \quad \left(\text{form } \frac{\infty}{\infty}\right) \text{ L'Hospital} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{2x} \quad \left(\text{form } \frac{\infty}{\infty}\right) \text{ L'Hospital} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{2} \\ &= \infty. \end{aligned}$$

4.5 Curve Sketching

Domain ; $x \in ?$ x intercept(s) y intercept(s).

Intercepts ; Evaluate $f(x)=0$ and $f(0)$

Symmetry : Evaluate $f(-x) \Rightarrow \underbrace{f(x)}_{\text{y-axis symmetry}}, \underbrace{-f(x)}_{\text{origin symmetry}}$

Periodic if $f(x+p) = f(x)$
 $\uparrow$
smallest positive constant

Asymptotes : Horizontal $\rightarrow \lim_{x \rightarrow \pm \infty} f(x)$

Vertical $\rightarrow \lim_{x \rightarrow a^{+/-}} = \pm \infty$

Decreasing / Increasing : $f'(x) > 0$ and $f'(x) < 0$

Exclude where $x=0$ from interval

Extremes (local) : Let $f'(x)=0 \rightarrow$ 1st derivative test
 $f''(x)=?$ for point : $(c, f(c))$

Concavity : $f''(x)=0 \rightarrow$ 2nd derivative test

Inflection Points : Where $f''(x)=0$, $f''(x)=?$ for $P(i, f(i))$